

Capstone Project: Deliverable 3 — SQL Database Design & Data Ingestion

1. Objective

The objective of Deliverable 3 was to establish a structured SQL environment for the Black Friday Sales Analysis project, load the validated dataset, apply appropriate data types and constraints, and verify the integrity of the migrated data before beginning analytical SQL queries.

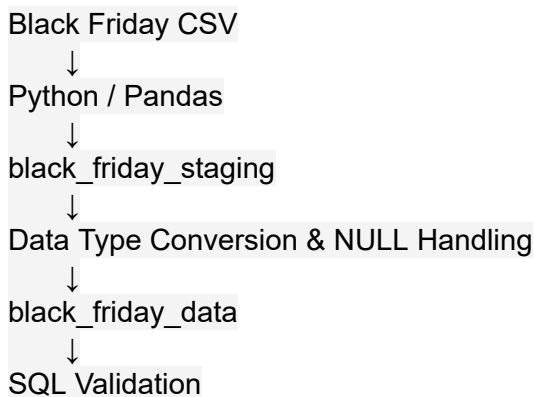
2. Database Setup

A MySQL database was created for the project under the schema name **black_friday_db**. Two logical layers were established:

- **black_friday_staging**: Temporary staging layer used to ingest the raw CSV data.
- **black_friday_data**: Final analytical table containing the cleaned and appropriately typed dataset.

3. Data Ingestion Process

Due to the size of the dataset (550,068 records), Python with Pandas and SQLAlchemy was used for efficient bulk ingestion. The data ingestion workflow followed this architecture:



The staging approach allowed the raw data to be loaded before applying the final database structure, data types, and primary key constraints.

4. Final Table Design

The final analytical table, **black_friday_data**, contains the following fields:

Column Name	Purpose / Description
User_ID	Customer identifier
Product_ID	Product identifier
Gender	Customer gender
Age	Customer age group
Occupation	Occupation category
City_Category	Customer city category
Stay_In_Current_City_Years	Duration of residence in current city
Marital_Status	Marital status
Product_Category_1	Primary product category
Product_Category_2	Secondary product category
Product_Category_3	Tertiary product category
Purchase	Purchase amount

Dataset Grain

One record represents a unique customer-product purchase transaction.

Primary Key Constraint

The composite combination of **User_ID + Product_ID** was validated as unique across all 550,068 records and established as the composite primary key.

5. Data Validation

Following data migration, SQL-based sanity and integrity checks were executed to confirm data completeness and accuracy.

Validation Metric	Result
Total Records	550,068
Unique Customers	5,891
Unique Products	3,631
Duplicate User + Product Combinations	0
Missing User IDs	0
Missing Product IDs	0
Missing Purchase Values	0
Negative Purchase Values	0
Zero Purchase Values	0